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URBAN GREEN SPACES AND THEIR IMPACT ON MENTAL HEALTH

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Abstract

Urbanization has significantly altered the environmental structure of modern cities by reducing the availability of natural landscapes and green environments. As urban populations grow, residents are increasingly exposed to environmental stressors such as noise pollution, overcrowding, and reduced access to natural environments. These conditions contribute to increased stress levels, mental fatigue, and other psychological challenges. Urban green spaces such as parks, gardens, tree-lined streets, and campus vegetation areas play a crucial role in promoting psychological well-being by providing restorative environments that reduce stress and support cognitive recovery. This research investigates the relationship between urban green spaces and mental health using a computational and data-driven approach. A framework is proposed that evaluates environmental resilience through a metric called the Context-Aware Green Resilience Index (CGRI). The framework integrates environmental parameters including vegetation coverage, noise levels, shade availability, crowd density, and duration of exposure. Vegetation coverage is estimated using computer vision techniques, while environmental stress indicators are incorporated to represent environmental pressure factors. The CGRI metric combines environmental capacity and stress pressure indicators to estimate the mental health resilience potential of green environments. An interactive analytical dashboard developed using Streamlit enables real-time visualization of environmental conditions, statistical analysis of resilience distributions, and environmental optimization recommendations. Experimental analysis demonstrates that environments with higher vegetation coverage and lower environmental noise levels exhibit significantly stronger resilience potential and contribute more effectively to stress reduction. The proposed system highlights the importance of integrating artificial intelligence and environmental analytics in urban planning to support healthier and more sustainable urban environments.

1. Introduction

Urbanization is one of the most significant global trends of the modern era. Over the past few decades, cities have expanded rapidly due to population growth, economic development, and technological progress. While urban development provides improved infrastructure and economic opportunities, it also introduces environmental challenges that affect human health and quality of life. One of the most noticeable consequences of urbanization is the reduction of natural landscapes within cities. Urban expansion often replaces forests, fields, and natural vegetation with concrete structures, roads, and commercial buildings. This transformation reduces the availability of green spaces that provide natural environments for relaxation and recreation. Urban green spaces play an important role in promoting both physical and mental health. These spaces include parks, botanical gardens, urban forests, recreational fields, and campus green areas. Green environments provide opportunities for relaxation, social interaction, physical activity, and psychological restoration.

Mental health has become an important concern in urban societies. Studies indicate that urban residents often experience higher levels of stress compared to individuals living in rural areas. Environmental factors such as traffic noise, overcrowding, air pollution, and lack of natural environments contribute to increased psychological pressure. Environmental psychology research suggests that exposure to natural environments helps restore cognitive resources and reduce stress. Natural landscapes provide visual and sensory experiences that allow individuals to recover from mental fatigue and improve emotional well-being. Despite the growing recognition of the benefits of green spaces, urban planning decisions often prioritize economic development over environmental sustainability. As a result, many urban environments lack sufficient green spaces to support mental well-being. Most existing research on green spaces and mental health relies on surveys or observational studies. While these methods provide valuable insights, they often lack quantitative environmental measurements. This research introduces a computational framework that integrates environmental sensing, computer vision, and statistical analysis to evaluate the mental health resilience potential of urban green spaces.

2. Problem Statement

Rapid urban development has significantly reduced the availability of green spaces in many cities. The absence of natural environments exposes urban residents to environmental stressors such as noise pollution, overcrowding, and limited opportunities for relaxation. These environmental conditions contribute to increased levels of stress, anxiety, and psychological fatigue. Although several studies highlight the importance of green spaces in promoting mental health, there is a lack of computational systems capable of objectively evaluating the mental health benefits of urban green environments.

Most existing research methods rely on subjective perceptions or survey responses. These approaches may not accurately capture the environmental conditions that influence psychological well-being. Furthermore, environmental factors such as vegetation density, noise levels, shade availability, and crowd density interact in complex ways that affect the restorative

potential of green environments. Without quantitative analysis, it is difficult to determine which environmental conditions provide the most beneficial impact on mental health. Therefore, there is a need for a data-driven system capable of analyzing environmental characteristics of urban green spaces and estimating their potential contribution to mental health resilience.

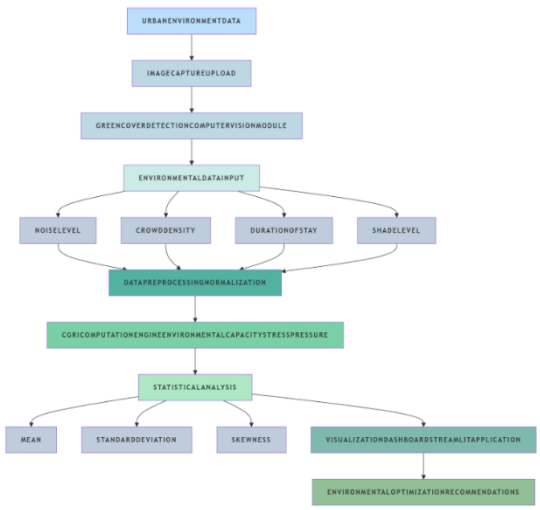
3. Objectives

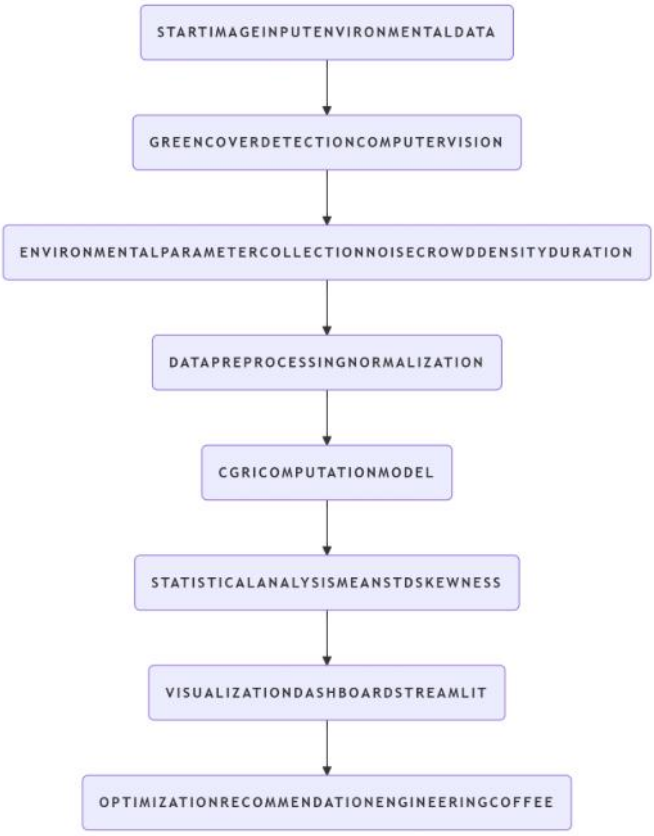
The main objectives of this research are:

- To analyze the relationship between urban green spaces and mental health outcomes.
- To develop a computational framework that evaluates environmental conditions of green spaces.
- To estimate vegetation coverage using computer vision techniques.
- To incorporate environmental stress indicators such as noise levels and crowd density.
- To develop the Context-Aware Green Resilience Index (CGRI) as a quantitative measure of environmental resilience.
- To design an interactive analytical dashboard for visualizing environmental resilience patterns and statistical distributions.

4. System Architecture

The proposed system architecture integrates multiple modules responsible for environmental data acquisition, image processing, resilience computation, and visualization.





The modular architecture ensures flexibility and scalability, allowing each component to be improved or extended independently in future research.

5. Methodology

The research methodology consists of several stages including environmental data collection, vegetation detection, data preprocessing, resilience computation, and visualization.

Data Collection

Environmental parameters were collected from campus environments and simulated datasets representing different environmental conditions. These parameters include vegetation coverage, noise levels, shade availability, crowd density, and duration of stay.

Vegetation Detection

Vegetation coverage is estimated using computer vision techniques. Images are converted from RGB color space to HSV color space to isolate vegetation colors more effectively. Threshold segmentation is applied to identify green pixels within the image.

Green cover percentage is calculated using pixel analysis:

$$\text{Green Cover \%} = (\text{Number of Green Pixels} / \text{Total Pixels}) \times 100$$

Noise Measurement

Environmental noise levels are estimated using microphone input and signal processing techniques. Audio signals are captured for short durations and processed to calculate the root mean square (RMS) amplitude.

The RMS value is converted into an approximate decibel scale representing environmental noise intensity.

Data Normalization

Environmental parameters are normalized to ensure that variables with different scales can be integrated within the resilience model.

Resilience Computation

Normalized environmental parameters are used to compute the Context-Aware Green Resilience Index.

6. Mathematical Model

The Context-Aware Green Resilience Index (CGRI) integrates environmental capacity and stress pressure factors.

Environmental Capacity (EC) represents positive environmental attributes.

$$EC = \frac{GreenCover + ShadeLevel + Duration}{3}$$

Stress Pressure (SP) represents environmental stress indicators.

$$SP = \frac{NoiseLevel + CrowdDensity + ExamPeriod}{3}$$

The final CGRI value is computed as:

$$CGRI = EC - SP$$

Where:

GreenCover = vegetation coverage percentage
ShadeLevel = tree canopy coverage level
Duration = time spent in the environment
NoiseLevel = environmental noise intensity
CrowdDensity = number of people in the area

Higher CGRI values indicate stronger environmental resilience.

7. Implementation

The proposed system was implemented using Python and several open-source libraries.

Component	Technology
Programming Language	Python
Dashboard Framework	Streamlit
Image Processing	OpenCV
Data Analysis	Pandas
Visualization	Plotly
Noise Detection	SoundDevice

The Streamlit framework was used to create an interactive web application that allows users to upload images, adjust environmental parameters, and visualize CGRI results.

8. Results and Discussion

Experimental analysis was conducted using environmental data collected from simulated campus environments. Histogram visualization was used to analyze the distribution of CGRI scores. Statistical measures such as mean, standard deviation, and skewness were computed to evaluate resilience patterns. Results indicate that environments with higher vegetation coverage and lower noise levels consistently produce higher resilience scores. These environments provide better conditions for psychological restoration.

Conversely, environments with high crowd density and noise exposure tend to exhibit lower resilience scores due to increased environmental stress. The analysis demonstrates that environmental characteristics play a crucial role in determining the mental health benefits of green spaces.

9. Conclusion

Urban green spaces are essential components of sustainable cities that support psychological well-being and environmental health. This research presents a computational framework for analyzing the mental health resilience potential of urban green spaces using environmental data and artificial intelligence techniques.

The Context-Aware Green Resilience Index provides a quantitative method for evaluating environmental resilience based on vegetation coverage and environmental stress factors.

The developed analytical dashboard demonstrates how computational tools can assist researchers and urban planners in designing healthier urban environments. Future research may incorporate larger datasets, satellite imagery, and geographic information systems to further improve environmental analysis.

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